Supplementary Information

S1. Complete paired official record and provenance

The table preserves all eleven paired observations. No fuel-intensity value was interpolated before 2015. The main comparison begins in 2017 because that bulletin documents a change in production-data processing and audit rules. Reconstructed fuel is not independently observed total consumption.

Year	Q (10^8 tonne-km)	I (kg/tonne-km)	Reconstructed fuel (Mt)
2015	851.65	0.294	25.038510
2016	962.51	0.293	28.201543
2017	1083.08	0.293	31.734244
2018	1206.53	0.287	34.627411
2019	1293.25	0.285	36.857625
2020	798.51	0.316	25.232916
2021	856.75	0.309	26.473575
2022	599.28	0.302	18.098256
2023	1188.34	0.292	34.699528
2024	1485.17	0.271	40.248107
2025	1640.83	0.260	42.661580

The canonical `source/verified_data.csv` contains separate source URLs, full SHA-256 digests, physical PDF page locators and the coverage interpretation for every pair. The official-source observation ledger adds exact Chinese quotations and source-file metadata. Transport-aviation interpretation is anchored to the CAAC 2019 transport-intensity statement, the 2022 policy/action series and indicator 126 in the 2024 survey system. This is a documented continuity interpretation rather than enterprise-level matching in every year. NBS cross-publication checks are retained separately; they reuse CAAC-origin statistics and are not independent original measurements. The 2025 observation was obtained from CAAC, not an unverified NBS 2026 yearbook.

S2. Chain-linked and direct endpoint decomposition

The chain calculation sums adjacent-year LMDI components. Direct endpoint allocation can differ, particularly through contraction and recovery; both sum exactly to the same endpoint fuel change. The following table reports Mt, retaining three decimals. All underlying calculations retain floating-point precision.

Period	Chain activity	Chain intensity	Endpoint activity	Endpoint intensity	Net
2017–2019	6.059	-0.936	6.071	-0.948	5.123
2019–2022	-20.843	2.084	-20.287	1.528	-18.759
2022–2025	29.931	-5.368	28.853	-4.290	24.563
2017–2025	15.147	-4.220	15.340	-4.413	10.927
2019–2025	9.088	-3.284	9.448	-3.644	5.804
2015–2019	12.846	-1.027	12.769	-0.950	11.819

Period	Chain activity	Chain intensity	Endpoint activity	Endpoint intensity	Net
2015–2025	21.933	-4.310	21.687	-4.064	17.623

The extended 2015–2025 comparison crosses the 2017 processing change and is not presented as a harmonised main estimate. Contributions remain accounting allocations; they do not identify technology, policy or rebound effects.

S3. Reporting-precision perturbations

At every combination of low/high starting and ending values, Q was perturbed by ±0.005 (10^8 tonne-km) and I by ±0.0005 kg/tonne-km. Sixteen corners were evaluated per period. This examines source rounding only, not systematic measurement error, population coverage or a statistical confidence interval.

Period	Fuel growth, central (%)	Fuel-growth corner range (%)
2017–2019	16.145	15.742 to 16.548
2019–2022	-50.897	-51.065 to -50.729
2022–2025	135.722	134.877 to 136.570
2017–2025	34.434	33.946 to 34.924
2019–2025	15.747	15.321 to 16.174
2015–2019	47.204	46.695 to 47.715
2015–2025	70.384	69.766 to 71.004

S4. Bayesian diagnostic, baselines and small-sample exclusions

Let d denote adjacent annual log-intensity changes. The conjugate model uses $\mu_0=0$, $\kappa_0=1$, $\alpha_0=2$ and $\beta_0=0.05^2$. For n changes with average $\bar{d}$, $\kappa_n=\kappa_0+n$, $\mu_n=(\kappa_0\mu_0+n\bar{d})/\kappa_n$, $\alpha_n=\alpha_0+n/2$, and $\beta_n=\beta_0+\Sigma(d-\bar{d})^2/2+\kappa_0 n(\bar{d}-\mu_0)^2/(2\kappa_n)$. The h-step log-intensity predictive distribution is Student-t with $2\alpha_n$ degrees of freedom, location $\ln(I_T)+h\mu_n$ and scale $\mathrm{sqrt}[(\beta_n/\alpha_n)(h+h^2/\kappa_n)]$. Exponentiation supplies a median and central intervals; the exponentiated Student-t has no finite positive moments, so an original-scale predictive mean is not reported.

B0 is unchanged intensity. B1 adds the unshrunk training mean log change. B2 uses the shrunk posterior location. Every model uses the same eligible origin, with a minimum of eight prior valid annual differences; no target observation enters its training. MASE divides absolute prediction error by the mean absolute adjacent-year naive error in that origin's training set. A zero denominator remains undefined.

Model	Test years	n	MAE (kg/tonne-km)	MASE
B0	2024, 2025	2	0.016000	1.894853
B1	2024, 2025	2	0.014655	1.750015
B2	2024, 2025	2	0.014790	1.764619

These two tests are diagnostics, not evidence of model superiority. The post-2017 series has no eligible out-of-sample test under the same rule. Excluding every training

difference touching 2020–2022 leaves only six valid differences in the extended record, so the pandemic-exclusion forecast is skipped rather than formed by joining separated years. Prior scales 0.02^2 and 0.10^2 are retained in the machine-readable outputs. The forecasting diagnostic is not used to choose or calibrate the conditional scenario domain.

S5. Conditional exploration, resolution and analytical bounds

The exact input configuration is `scenario_config.json`. g is uniform in the log-rate interval corresponding to ordinary growth of 2–6%; b is uniform in the continuous intensity-decline interval corresponding to ordinary reductions of 0–2%; θ and r have uniform design weights over 0.10–0.30 and 0.02–0.12, respectively. These are exploratory weights, not inferred parameter distributions. Delays are evaluated separately, not pooled into a probability mixture. H=15 and κ=1 define a 2040 non-increasing-fuel benchmark, not an asserted official target.

Delay (years)	2^{12} design share meeting target (%)	2^{14} share (%)	2^{14} terminal-ratio median
0.0	7.0312	6.7932	1.346867
3.0	5.2246	5.1147	1.370412
8.0	2.6367	2.6062	1.423835

The shares above describe only the stated design domain. They are not probabilities that China will meet a target. Scrambling seeds are 20261016, 20261017 and 20261018. No points are skipped or thinned. The higher-resolution results are stored with the resolution comparison; the initial individual design points are supplied in CSV. Re-running the documented code regenerates both.

By monotonicity, the lowest terminal ratio occurs at low g, high b, high θ, high r and the shortest delay; the highest occurs at the opposite corner. Exact bounds are 0.739640 and 2.364954. These are bounds for the assumed box, not empirical prediction limits. A cumulative-fuel diagnostic sums future years 1 through H, excluding the observed baseline year, in units of baseline annual fuel. It is retained in the numerical output but is not used to claim a carbon budget.

The analytical frontier explicitly distinguishes baseline-sufficient, full-adoption-insufficient and deployment-threshold cases. It does not truncate an impossible adopted share to one or interpret fuel-use-weighted adoption as aircraft-count replacement.

S6. Reproduction and integrity boundaries

The replication README gives the offline reproduction command, exact software versions and input hashes. Eighteen deliberately synthetic arithmetic/mathematical checks cover accounting identities, unit conversion, Bayesian conjugacy and prediction variance, forward-only training, missing/excluded-year handling, adoption boundaries, analytic/numerical threshold agreement, monotonicity and exact scenario corners. These tests are not empirical observations. Official-document hashes, table transcriptions, coverage reasoning and figure-source values are separate evidence checks.

References were checked against DOI metadata and authoritative publisher records. Reading coverage differs by source: the Bergero and Dray papers were used at the

official-abstract level; Ang (2005) was bibliographically verified but its full text was not retrieved. The decomposition equations and closure checks are self-contained. Current MDPI author instructions, scope and research-data/artificial-intelligence policies were read; these checks do not constitute a journal acceptance decision or human-author attestation.

No confidential thesis survey responses or unverified capability scores were reused. Public archive publication and formal submission have not been performed. Author details, contributions, funding, conflicts, prior-thesis relationship and human-author review remain to be completed by the authors.